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Case Study: Cutting the Peak of a 56 kW Heat Press with a VFD and Heater Power Control · Ban Doo Chiangrai

Rev01 · 10 November 2026 · Evidence: SolisCloud 5-minute data (29 Oct – 8 Nov) · Heavy-Machine Days Report · VFD Advice Rev02 · Integrated Advice Rev01 · six PEA bills · PEA tariff 3.1.3

SolisCloud data confirms the **55–62 kW peaks come from the drum heaters**, not a motor. We propose a **small VFD for the drum motor**, together with an **SCR heater power controller with PID control** and a new warm-up schedule. This lowers the billed peak from **≈ 61.5 kW to 32–38 kW** and cuts the electricity bill by about **33–46%**.

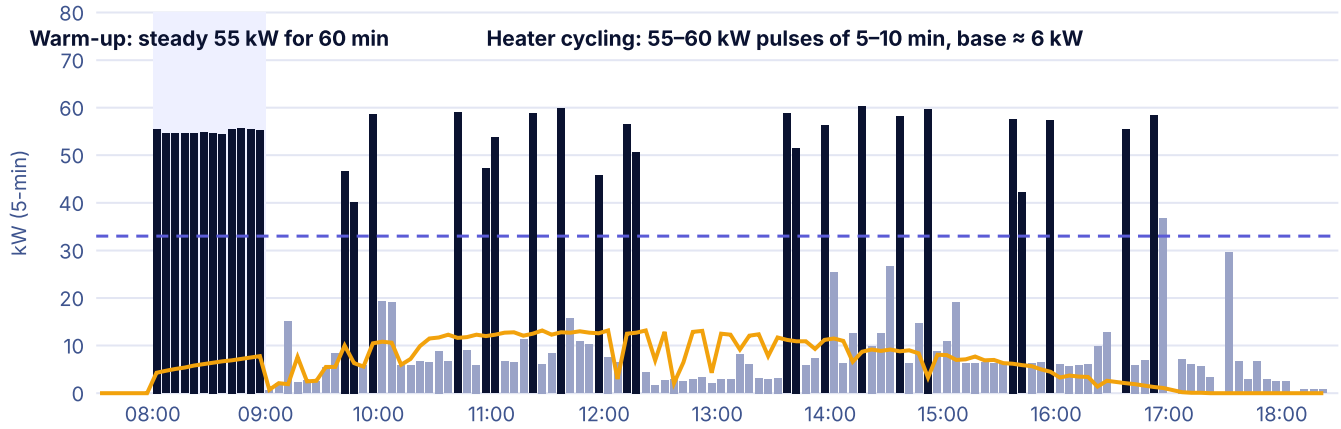
Billed peak
61.5 → 35 kW
range 32–38 kW with the control package

Monthly bill (estimate)
−33 to −46%
full package; scheduling alone −21 to −34%

Warm-up
55–61 kW
for 55–60 min; it sets the whole month's peak

1 SolisCloud evidence: machine days

31 Oct, 5-minute data: steady warm-up, then heater pulses



Load > 40 kW (heaters on) · Other load · Solar output · Target ceiling ≈ 33 kW (dashed)

Source: SolisCloud, Ban Doo Chiangrai plant, 5-minute data, Saturday 31 Oct 2026.

Day	Use (kWh)	Warm-up	15-min peak	After warm-up
Sat 31 Oct	213 (solar 72)	08:00–08:55, 60 min, 55.0 kW	55.4 kW	22 pulses until 16:50; highest 15-min 37.8 kW
Wed 4 Nov	157* (solar 42)	08:10–09:00, 55 min, 55.1 kW	56.2 kW	16 pulses until 16:20; highest 15-min 41.6 kW
Sat 7 Nov	240 (solar 38)	11:55–12:45, 55 min, 58.1 kW	60.9 kW	25 pulses until **21:40**; highest 15-min 47.4 kW; 100 kWh after 17:00

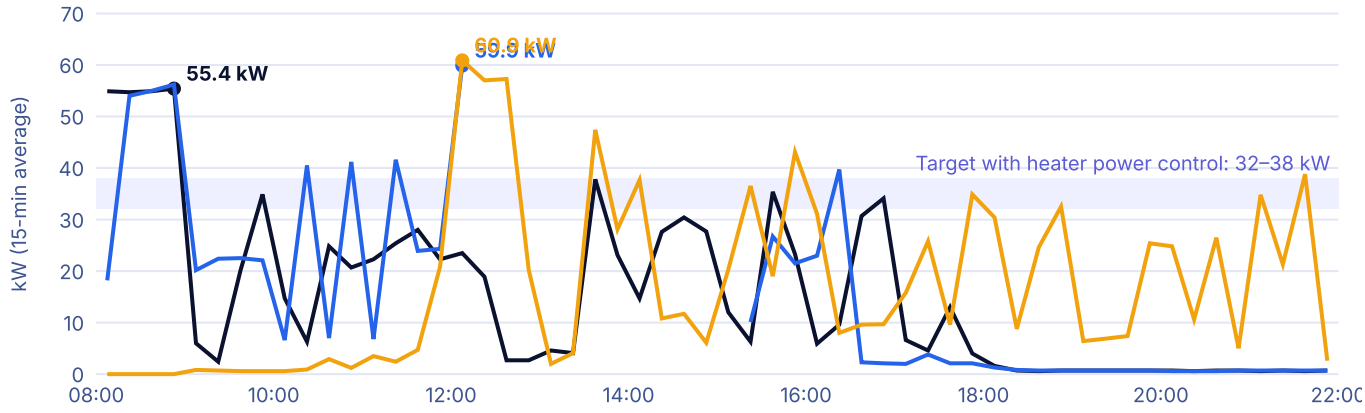
Normal day	40–58	none	8–12 kW	Smooth 2–8 kW load

* 4 Nov has missing data 12:00–15:25. Energy figures are computed from the 5-minute data.

2 What the evidence shows

Evidence	Conclusion
Flat 55–58 kW for 55–60 min on every machine day	The cold drum is heated at full heater power. A flat draw like this is a resistive load; a motor does not behave this way.
55–60 kW pulses of 5–10 min; ≈ 5–6 kW base between pulses	On/off temperature control switches all heaters together (heaters on 16–32% of the time).
Each day's highest 15-min average (55.4–60.9 kW) falls in the warm-up	The warm-up sets the month's demand charge.
Fixed part of six PEA bills	Matches the tariff 3.1.3 demand charge (221.50 THB/kW) at ≈ 61.5 kW.
7 Nov: heaters still cycling at 21:40	The drum is kept hot long after sunset, all on grid power.

15-minute averages (PEA's billing basis) on machine days



Every day's highest point comes from the warm-up. After warm-up, overlapping pulses still push 15-min averages to 38–47 kW.

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3 Proposed equipment

Item	Specification	Purpose
A. Drum motor VFD	3-phase 380 V, Heavy Duty, one size above the motor (e.g. 1.5 kW motor → 2.2 kW VFD; 2.2–3 kW → 4 kW), sensorless vector, with STO. Example ranges: Delta MS300, Schneider ATV320, ABB ACS380, INVT GD20.	Sets drum speed to the required heat dwell time and gives a soft start. Install only if the machine has no speed control.
B. 3-phase SCR heater power controller	≥ 100 A (≈ 85 A × 1.2), burst-fire/zero-cross, power-limit input, semiconductor fuses, heatsink and fan.	Caps heater power at 50–60% (≈ 27–33 kW) during warm-up, and supplies steady power instead of 55 kW pulses.
C. PID temperature controller	4–20 mA or 0–10 V output to the SCR, with ramp/soak for a staged warm-up.	Holds drum temperature with low, even power.
D. Peak alarm (optional)	Energy meter with a 15-min demand alarm, or alerts from Solar ESS monitoring.	Warns before the peak passes the set ceiling.

> **Caution:** most heat presses must keep the drum turning while hot so the felt blanket does not burn. Set the VFD to keep turning after a stop command and wire its fault contact as the machine maker specifies. A 75 kW VFD or a 55–75 kW soft starter is not recommended for this machine.

4 New machine-day routine

08:30–09:00
Start warm-up at limited power, ≈ 30 kW

→

10:15–10:45
Drum ready; solar supplies 8–13 kW

→

10:30–15:30
Continuous pressing; PID keeps power even

→

By 16:00–17:00
Heaters off or at setback; no cycling into the night

Warm-up energy is unchanged (≈ 50–55 kWh) but spread over about 1.8 times as long, so it no longer creates a peak.

5 Expected results

Monthly saving	A. Scheduling only	B. Full VFD + SCR	Basis
Billed peak	38–47 kW	32–38 kW	from ≈ 61.5 kW
Demand charge	16–25% of the bill	25–32% of the bill	≈ 237 THB per kW removed (incl. VAT)
More solar used + phase balancing	200–350 kWh	250–500 kWh	Work moved into 10:00–15:00
Heaters off on time, no evening cycling	100–170 kWh	100–170 kWh	7 Nov: 100 kWh after 17:00
PID + insulation + drum VFD	—	90–200 kWh	Less overshoot and heat loss
Total bill reduction	**21–34%**	**33–46%**	≈ 3.57 THB/kWh incl. Ft and VAT

6 Conditions for the savings

- **One peak sets the whole month:** a single full-power warm-up (e.g. a cold start at 08:00 on any one day) cancels the month's demand saving. The SCR limit enforces the cap automatically every time.
- **Minimum charge:** PEA charges at least 70% of the highest demand charge in the last 12 months. This site's new bills stay above that floor, so the full saving applies from the first month.
- **No other large loads** during warm-up.
- **Confirm the tariff** (3.1.3) and meter size from an actual bill before final figures.
- **Measure the result** on SolisCloud every machine day and compare with the next bill.

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7 Implementation steps

1. **Week 1:** safety check of the machine supply (85 A rated) and start the new warm-up schedule straight away (package A).
2. **Weeks 1–2:** get heater groups, temperature controller, drum motor and speed-control details from the machine maker.
3. **Weeks 2–4:** quote and install SCR + PID and the drum VFD (if needed), by the owner's electrician or the machine maker.
4. **Weeks 4–5:** set the power ceiling, test warm-up time; Solar ESS checks the result on SolisCloud.
5. **Following month:** compare the PEA bill with the previous month and report the saving.

Work on the machine and the building wiring is inside the owner's installation. It needs the owner's consent and is carried out by the owner's electrician or the machine maker to Thai EIT standards. Solar ESS gives technical advice, monitors results and adjusts the solar system to match. All figures are estimates from monitoring data.

Proposals welcome: Solar ESS invites proposals from experts to install the VFD and heater power control, and from consultants to lower the site's energy cost, at Ban Doo, Chiang Rai. Details and how to apply: www.solaress.org/job.html

Solar ESS Working Group